Two State Model Example

Two States

s= +1 ——————— *Energy =* ε

s= -1 ——————— *Energy = 0*

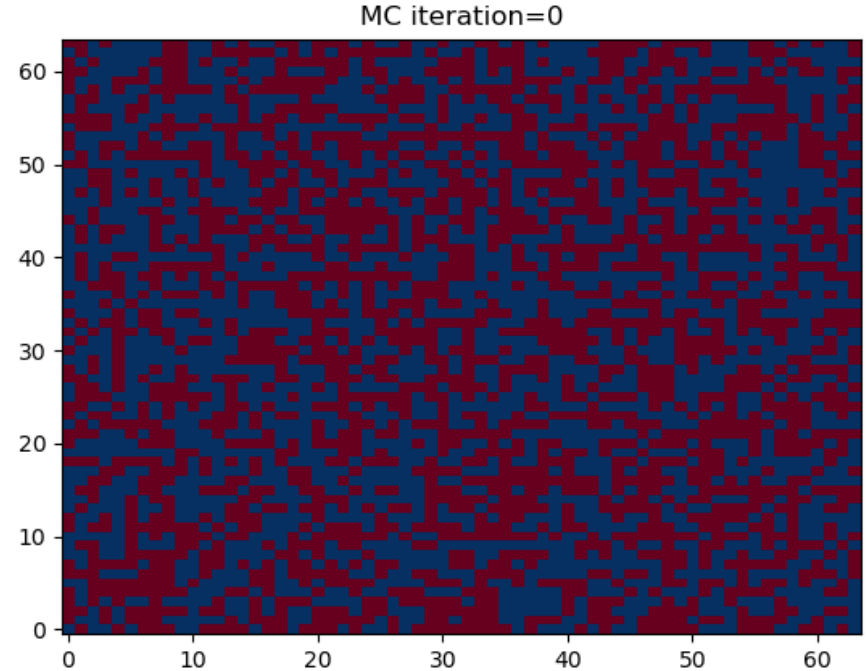
At a given T ***a Metropolis simulation*** *gives:*

- *State of the system (occupancy of each state)*
- *Energy of the system*

Python Available at our GitHub

https://github.com/jfaraudo/Monte_Carlo_Metropolis_Examples

Monte Carlo – Markov Chain Algorithm



Arbitrary initial condition:

Consider an initial state (*o*) with N sites (atoms) at random state (s=-1 or s=+1) => energy U(o)

1) Try a new state (n) with energy U(n):
Randomly flip one site

2) **Decide to accept or reject the change**

If $U(n)-U(o) < 0$ => accept the move from *o* to *n*

If $U(n)-U(o) > 0$ => accept the move from *o* to *n* with a probability given by the Boltzmann factor:

$$p(o \to n) = \exp\left[-\frac{U(n) - U(o)}{k_B T}\right]$$

- <u>Repeat over and over again</u> until a fixed number of steps or some convergence criterion is achieved

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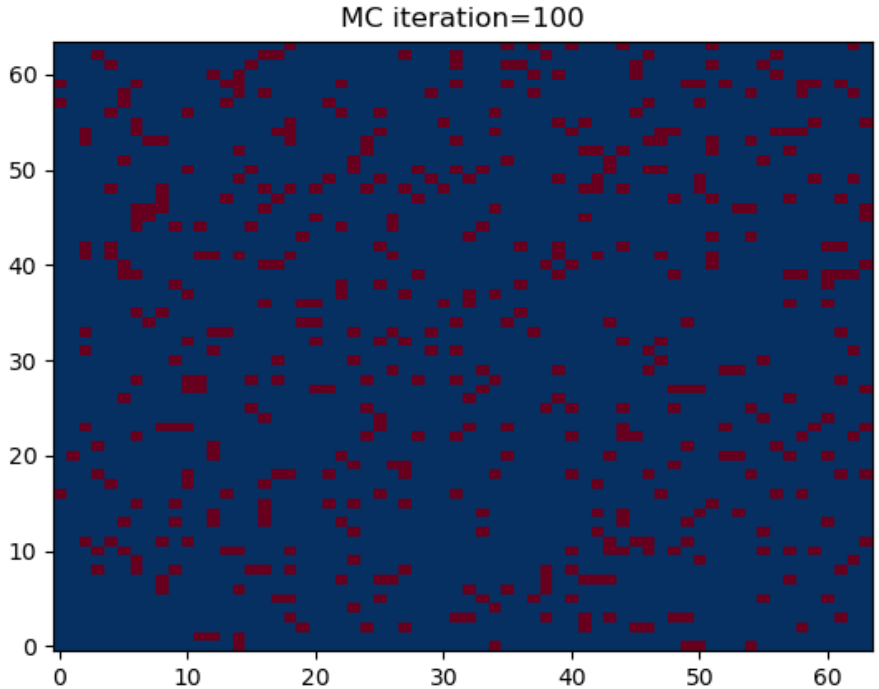
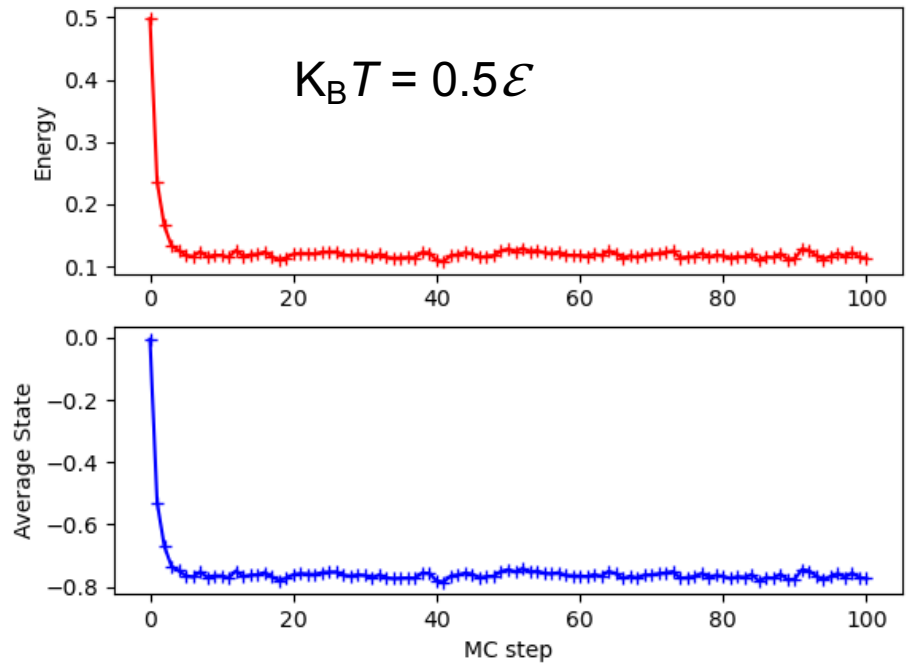
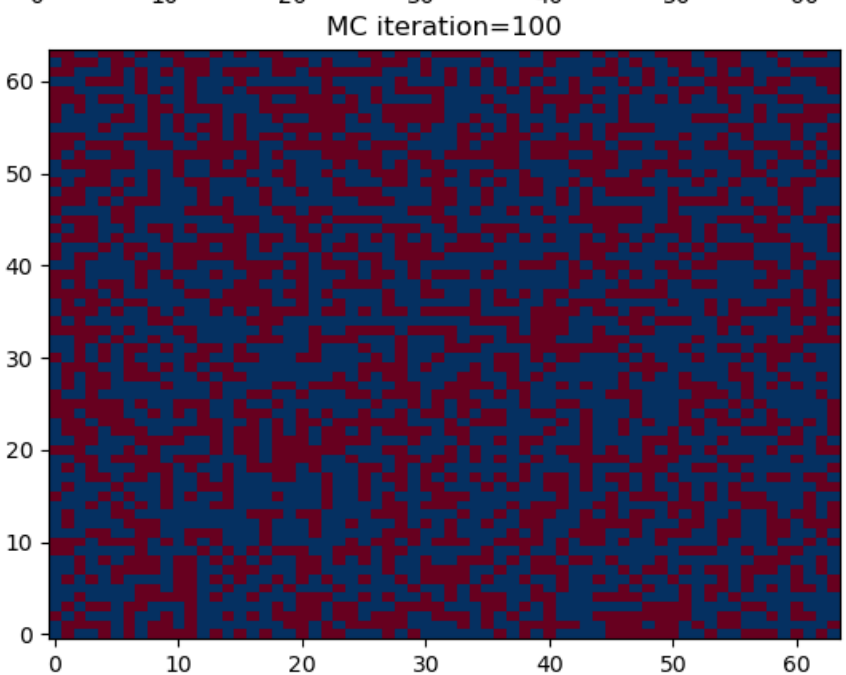
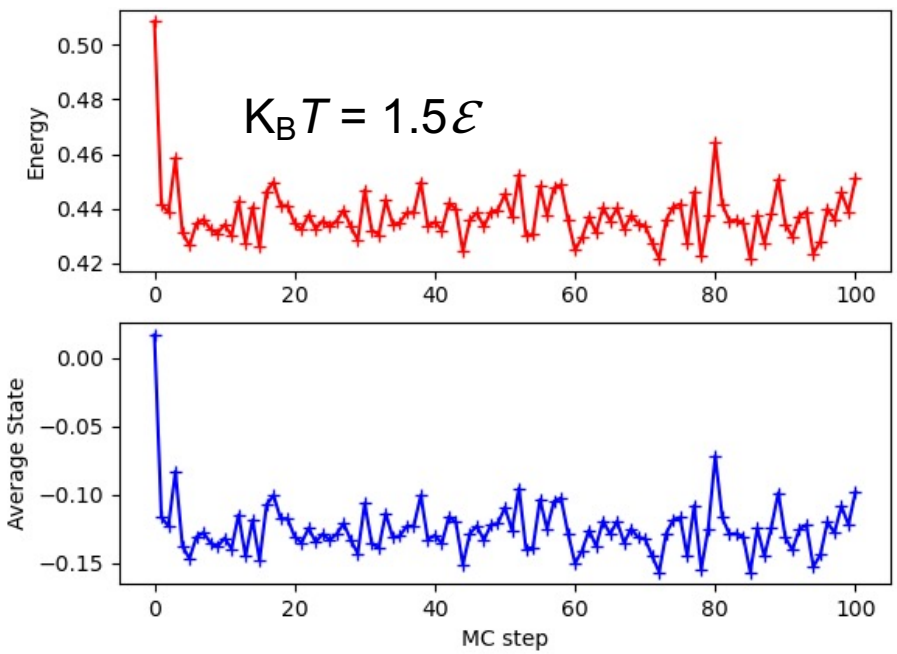
Typical Simulation results: https://github.com/jfaraudo/Monte_Carlo_Metropolis_Examples

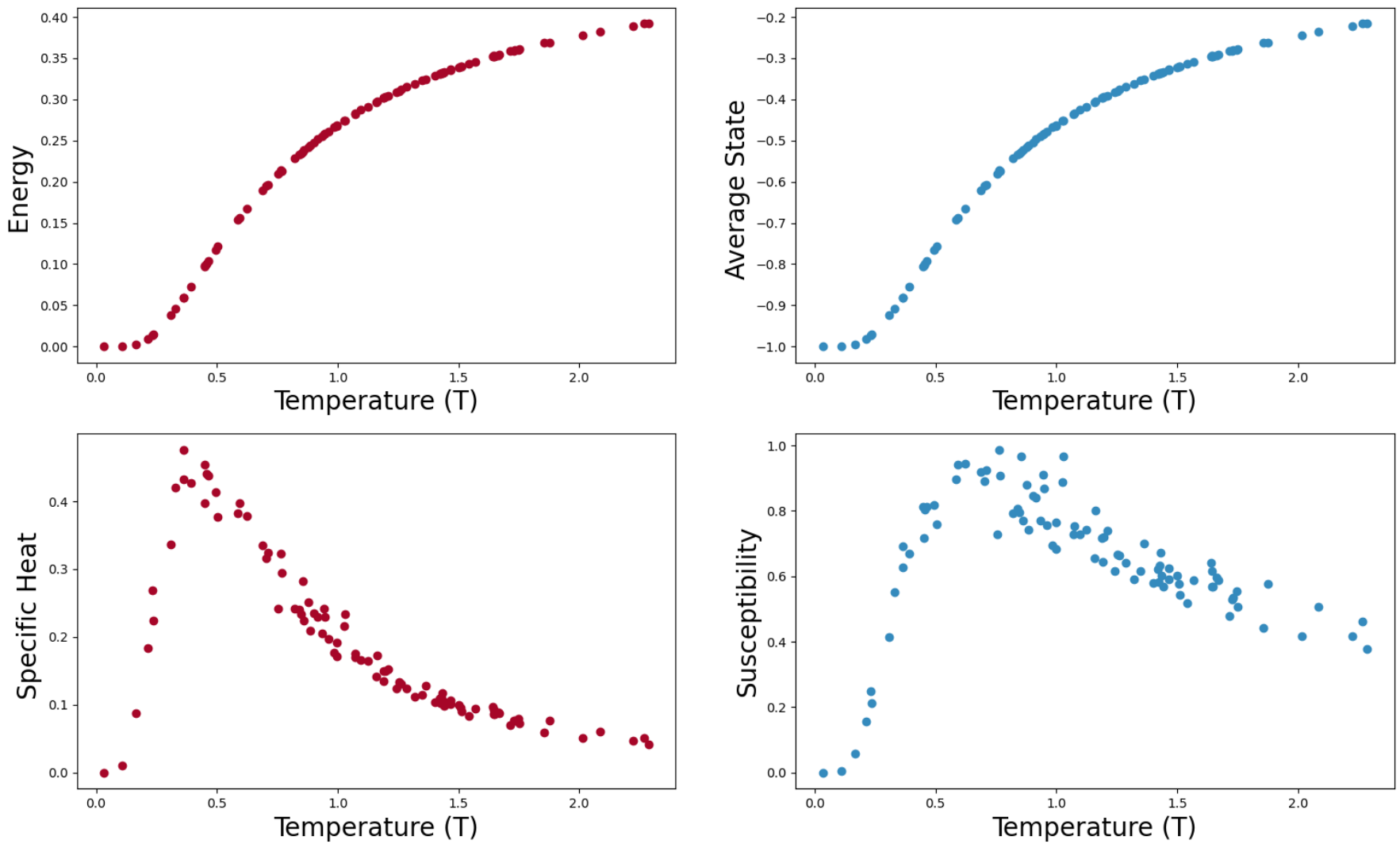








Simulation results: Dimensionless Temperature. $T^* = K_B T / \mathcal{E}$



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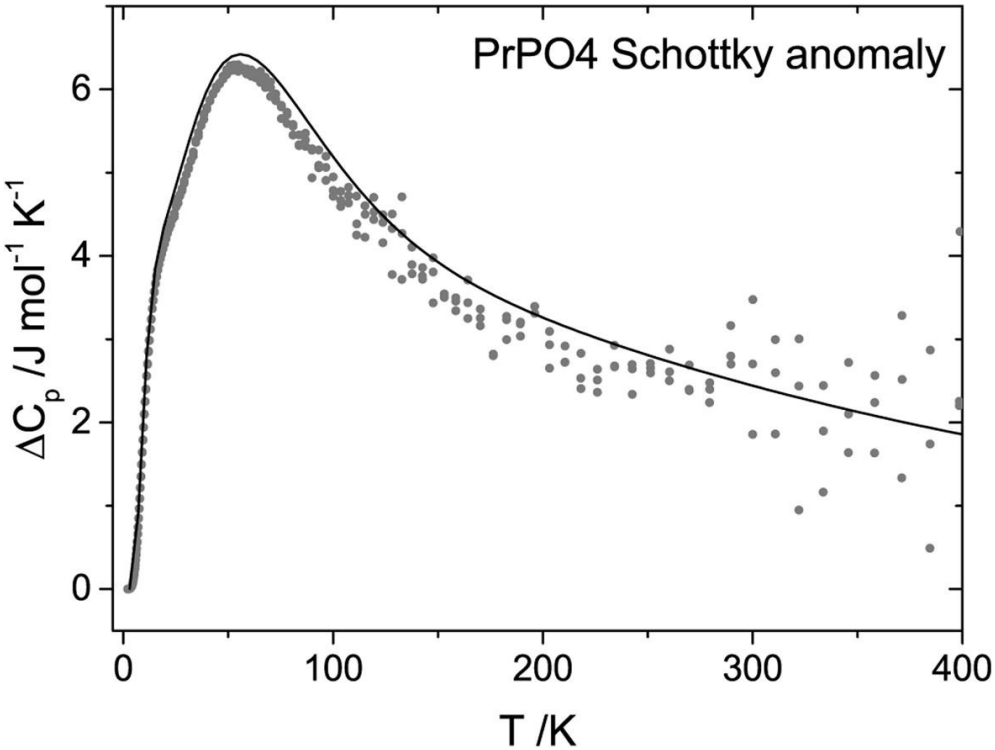
Examples of Two State systems:



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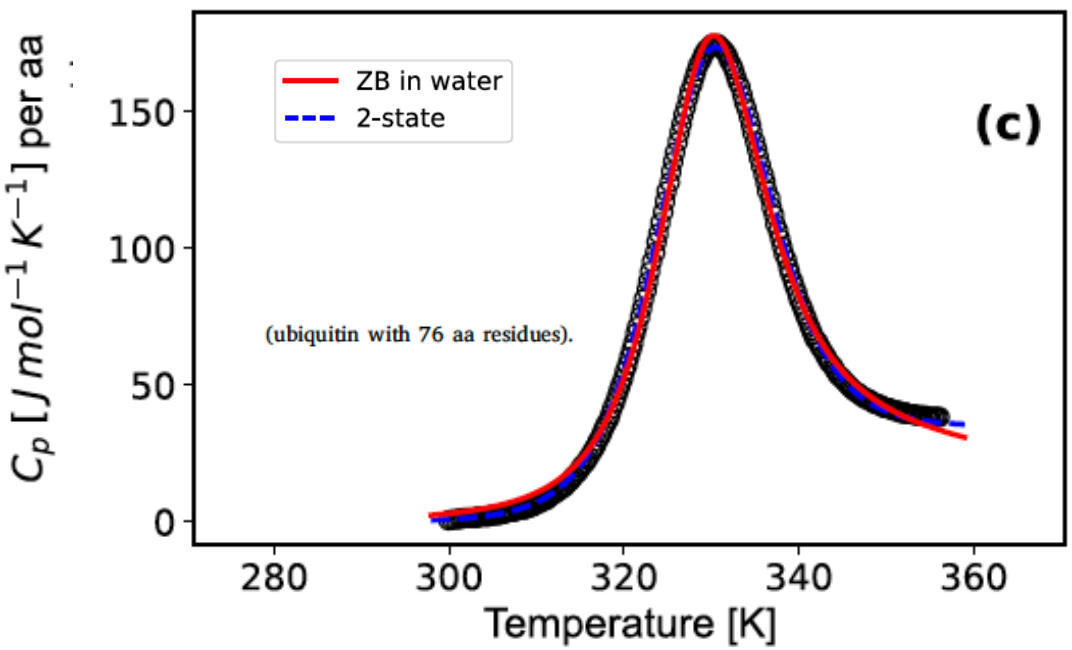

Heat capacity of proteins
Archives of Biochemistry and Biophysics 760 (2024) 110132
https://www.sciencedirect.com/science/article/pii/S0003986124002546?via%3Dihub